

ActInf Textbook Group ~ Cohort 8 ~ Session 6

(Chapter 3) ~ 3/27/2025

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appreciate all right hello welcome to the active inference institute's textbook group uh this is cohort 8 we are still in part one of the active inference textbook looking at chapter 3 we know from before that we saw the introduction to the active inference textbook and following that was chapter two starting from the so-called low road to active inference as the authors have provided us not one but two paths at arriving at active inference the low road started with basis theorem uh which is a classic uh formula for uh essentially updating one's beliefs in the sense of we already start with some prior beliefs we then encounter some new evidence or observations and then through relating uh our beliefs with observations we then can engage in what we would call inference and just carry out basis theorem in order to arrive at a new belief or a posterior belief so from there we can start to imagine entire generative models uh that have um that essentially organisms whether they be agents or however we want to conceive of them but some kind of generally autonomous or or distinguishable thing uh can have a generative model whereby it carries out basis theorem in varieties of ways including in very important uh uh behaviors that relate to adaptation um or otherwise such as gaining information about one's environment in order to update one's beliefs uh order to better uh uh navigate that environment and then we had notions of predictive coding we had planning as inference perception as inference um perception in one sense can just be viewed this is me putting it in caricature almost like a likelihood Matrix in the sense of we have observations and as long as we have some mapping uh between those observations and hidden States we can arrive at updating our beliefs and this is more or less the notion of perception as held not just an active inference but a variety of theories and cognitive science um again active inference is not necessarily Reinventing the wheel instead it is a normative framework that draws from a variety of fields in order to construct what is an attempt at a much broader coherent meat framework as as opposed to perhaps some of the more disparate theories that

currently exist um and so then we also have planning as inference in the basian brain now we'll be starting from the high road to active inference starting with the free energy principle now what is the free energy principle and why do we care about it and why does it have a nearly M magisterial sound about it whenever we hear it out of context well um with chapter 3 we start with this idea that uh as the authors claim uh active inference starts from first principles in statistical physics and the central imperative that organisms must maintain their existence so uh that is they must avoid surprising States and then introduces from there in the chapter the minimization of free energy as a computation tractable solution to this problem so it's not simply that the free energy principle is being stated um cely as a a kind of mechanical logical necessity it's actually in the service of something whenever we view it from the standpoint or or respective to uh organisms or or agents within their environment that includes looking at concepts of s self-organization of those agents um how they minimize their surprisal to remain within so-called preferred States what might be likened to the notion of homeostasis we have the idea of a marov blanket um which we may have witnessed before in this um part of the book but it is laid out much more clearly for us in this chapter over and above the other chapters um this is a notion that is very much worth getting familiar with um there are of course debates here around them uh what a Markov blanket is what it should entail um and there is also an important distinction between the marov blanket as defined by uh uh probability and causal inference theorist Judea Pearl uh as opposed to how this idea is employed in active inference um but nonetheless there are very strong uh sort of similarities between the two concepts and I would like to spend just a moment talking about Markov blankets just because they're so essential um to active inference and not just active inference but any kind of of of computational modeling right um one quick way of saying uh getting at the idea of a marov blanket is how do we Define a thing and and it's initially seems like a silly or strange question but it becomes really important whenever you want to engage in computational modeling for example if you're going to model the behavioral of a population of of neurons in some particular area or region of the brain you have to be able to figure out what are the neurons what are their boundaries and what is not the neuron and so sometimes I've used this this phrasing that I borrowed from another theorist Mark SS who is a neuros psychoanalyst who's done a lot of work with with Carl friston in active inference in the past where he he says there's the system or system of interest such as the organism and then there's the not system right we have to figure out what the the organism is not in order to understand it in its environment so the the environment is another word for the not system in this case and

so with a marov blanket this uh the the the author's quote an important precondition for any adaptive system is that it must enjoy some separation and autonomy from the environment without which it would simply dissipate dissolve and thereby succumb to environmental Dynamics and so unquote it is through surprise minimization that a system or organism or agent maintains this separation from its environment so we can see um for example in box 3.1 there is a somewhat clearer understanding supplied to us about what a marov blanket is but essentially if we can identify something like blanket States within a system then we can start to figure out how here they use μ which could for example represent the internal states of the agent versus X which might be a kind of true state of the environment by being able to define the blanket States we can begin to separate them out that brings us to figure 3.1 and so uh kind of classically in active inference the way the phrasing is that we have the internal States over here again defined by μ of the organism and then we have the external states with X but between them we have active States and sensory States these both it can be slightly confusing the way they they've placed blanket States down here but it's actually the active States and sensory states are both uh considered to be blanket States between and which essentially separate the external and internal States and so they phrase it as um a blanket for a variable can be identified by taking into account the variables parent variables which it depends on the variables child variables which depend on it and in some cases the other parent variables of its children variables what what is this about about parent and children variables for for those of us who are not uh fully attuned to sort of the the the statistical terminology here well we have let's say we're looking at the internal states of the organism well what do they depend upon if we look at the arrows here we see the sensory States the sensory states are being fed into the internal States What's Happening Here is perception the agent receives through its sensorium sensory States or observations right so it's kind of inheriting in a I want to be clear that inheriting is being used Loosely here but it's kind of inheriting aspects of the sensory States right it's taking them in and relating them such that through its internal mapping or likelihood Matrix is one way of phrasing it it can then infer uh its internal States for example uh sensory I notice my stomach is growling internal States in meaning beliefs I now believe oh I must be hungry my if my stomach is growling I must be hungry probability of hunger conditional on on observation here in a matrix might be incredibly high and so it leads me quickly to make that inference from there I could engage in let's say some kind of planning or policy inference such that I end up deciding what to do about this by determining what to do and then actually carrying out the action well now my body is acting in the environment I am effectively kind of

putting my my the the the outcome of my my inferential processes into the reality around me so now that becomes its own blanket state right is the action of me picking up you know a piece of food to eat it I mean that's happening outside of me right um and and so that's that's another thing that separates me yet vicariously connects me with the environment around me such as the piece of food um so that that's one way of of of kind of phrasing what a Markov blanket is and it's incredibly important when looking at at modeling because we we frequently think about as we'll see later in chapter six um whenever we want to start creating our own model and and and simulating things or simulation we have to very carefully specify what are these apparent internal States um whe whether we know their values at first or not we have to identify what they are what is it we're trying to model and then what is not and then what is it that connects them right so that's that's a sense of the the Markov blanket in a nutshell furthermore it kind of gets it something really important that we see whenever they start to illustrate some examples of this which is we've seen before uh the notion that uh an agent especially in Chapter 2 might start to synchronize in some way with its environment um which can lead some people astray and thinking that the generative model necessarily needs to somehow equate to the environment around it as if the internal States need to be perfectly identical to the external States um but I'll refer back to the kind of earlier quote an important Dynamics without meaning to sound too crude or or strange here I mean the environment around me doesn't care so much about my caloric nutrition and and you know none taken emotionally that's just how things work I'm the one who cares about that or in another sense it's my my body and the the physics and Physiology beneath it that cares about that if I gave in to the environmental Dynamics where my caloric nutrition simply runs out I will certainly succumb to the environment around me in some ways here I have to actively act against a natural tendency for my body to run out of caloric nutrition uh you know and as its gluc close levels start to deplete um so I have to do things in order to sort of resist that entropy and we see some of that kind of more um physics based language in in this chapter as well and thinking about it and thinking about hamiltonian uh physics and principle of least action so with figure 3.2 two and I'll I'll kind of maybe start to open things up not too long after this but the idea of figure 3.2 here is that we can see if we recall that X um for example could represent the the so-called true states of the environment so X is the the states of the environment and then we have μ over here which might the be the internal states of say a an agent or organism then we can also have the relationship between those things or the joint relationship or distribution um between those things and the blanket states that through

through which the these two different kinds of states the the internal and external are are vicariously related and so if we see for this blanket uh X that the environment actually has a positive correlation but μ actually has a negative correlation with that blanket and yet we do see a kind of synchrony uh here an inverse synchrony a negative correlation between uh this is supposed to be yeah it's labeled uh X and μ right so this despite the fact that we can kind of statistically separate X and μ they still have a rather you know generally I mean of course we' want to you know we could compute the P value or otherwise and and and or or look at a Pearson's coefficient or however you would like but essentially we can we can generally make out that there's a a negative correlation here so you know it's my body runs out of caloric nutrition that is a natural thing that happens in the environmental Dynamics in which I find myself um and yet um I have a tendency to act against this by bringing my Hunger down by eating and so we see that there is this kind of negative correlation between you know me losing caloric Nutri losing my energy and at the same time trying to fight against it right it's just to bear in mind again an example showing that there is no necessity that our internal States and I I keep phrasing it that way but there's no necess necessity that any thing or agent or organism has internal states that necessarily are one to one equivalent to environmental uh external States and in fact given my examples here that might actually be very dangerous if they were fully aligned such as me no longer eating food in order to you know get kind of give into the environment around me um so we we beyond that we're looking at at things like posterior beliefs like so how do we fight against this this natural tendency towards entropy or or more clearly interpretable definable like causal relationships we see such as you know the relationship between hunger and consumption um we the idea is that we engage in free energy minimization that is for those who are familiar with you know machine learning deep learning or otherwise we can view free energy minimization in part as a kind of optimization problem uh where we are essentially Trying to minimize a uh upper bound on surprisal surprisal in this case for an organism like me you know again this kind of semi- silly example of me eating food like um you know I need x amount of calories per day right if I eat too much above or below that I start to kind of fall outside of a generally uh a healthy uh caloric intake level or range of of caloric intake uh you know with with obviously very different consequences if I eat too much versus too little um but nonetheless it there's some kind of homeostatic range here that can be defined so it's kind of that homeostatic range that in one sense defines me as a thing in a physical environment and how I persist which is if I can maintain these various you know potentially very very many depending on how granular we want to be um you know

there could be very many um U preferred States or homeostats that I need to maintain it could be blood oxygen flow it could be caloric nutrition it could be simply you know my ability to engage in the world around me in order to uh maintain an income uh in a market economy in order to you know get the means for my survival in the first place um and so we have that and then we have surprise meaning it's almost like the the inverse it'd be like dispreferred States we don't want surprise right I don't want my caloric intake to be too low um and I don't want it to be too high I especially don't want it to be too low um from there what what this does whenever we look at this in like a basian framework is that this opens up the idea of we have expectations and leaves and so I might a way we could phrase this is my body expects to realize its preferences or its homeostatic ranges uh in practice and so there's a kind of belief or expectation that I will remain within my preferred homeostatic ranges um in this sense uh we are self-evidencing organisms would be self-evidencing because the idea is this is a beliefs based framework we can phrase it that way they in fact believe to be uh you know with a very high probability within their respective homeostatic ranges um this of course can lead to a variety of interesting kind of Dynamics um whether you're familiar with machine learning or not uh you you can you know many people can int it the idea of optimism bias or the idea that someone might have such a strong belief in in something in particular that they will bring it about or they will just assume it is true and they might even be opposed to uh disconfirming evidence you know evidence to the contrary of what they believe um another notion that relates to this is uh the phrasing uh hyper precise priors which is very relevant here in a basian context as we already know that priors or the kinds of beliefs that you bring to a situation whether they be beliefs about um in states of the world but also actually have beliefs about prior about observations themselves which are usually used as the kind of homeostatic ranges for example I expect to not see my stomach growling um if it's the case that it's not growling then that can also ensure that I'm not hungry or another way of phrasing it is to not be hungry is to not see my stomach growling and if I prefer not to see my stomach stomach growling and that's a that's a kind of sign that all is well right meanwhile with Hyper precise priors that would be the idea that you have some kind of um quote unquote excessive excessively strong prior beliefs about something that uh might be might be so terribly strong and and so terribly kind of precise or certain that it is very difficult to um start to disbelieve or unlearn uh uh uh particularly what might be harmful beliefs right so this can be used in the study of say computational Psychiatry or pathological Behavior Uh including looking at something like delusional uh thinking uh in let's say schizophrenia where one way we could phrase it is there

are such strong priors about uh what is occurring essentially in one's own mind that they kind of overtake one's perception of the reality around them which results in this kind of very intensive hallucinatory behavior when of course the hallucinations are being produced by what we might phrase as the the priors of the mind um of course none of this at all is assuming that the person is intentionally trying to make any of this happen or intentionally choosing to believe in this way it's not necessarily a soft belief um in the sense of you know oh I I choose this out of free will it might be that you know it's a belief spased framework it might be that genuinely computationally neuronally or otherwise uh there there's something like priors that are just excessively strong that lead to the kind of collective neuronal outcome that is production of hallucinations so in any case what do we do we we we self-evidence that is we we seek out information that actually reconfirms our beliefs or our expectations about ourselves and the world around us um this you know for those who are interested in in kind of you know social science or or other fields like that that would also imply that you know actually a lot of these concerns about about um you know opinion polarization or Tendencies for people to form in groups and outgroups in order to just maintain their own respective social bubbles um and opinion that that are very polarized between themselves versus relative to other groups um there is you know a lot of kind of theoretical and and empirical support for those kinds of Dynamics and and we very much find them in in active inference um in in fact a really nice um paper in the past that I've always found very interesting was uh this one where they actually built a full active inference simulation uh epistemic communities under active inference with some recognizable uh researchers such as Conor Hines Maxwell ramet barison and so they they construct a model that essentially looks at confirmation bias Conformity um ties together a lot of otherwise what might be potentially loose threads in uh psychology a as is you know what has been viewed as predominantly a theoretical field uh rather than necessarily like strictly a quote unquote hard science and yet this um the kind of modeling involved here and Reliance upon you know physical and and neuroscientific principles does allow us to start to actually formalize maybe this is not the best image to start with just because of the complexity uh but um to start to formalize you know these kinds of Dynamics in interesting ways where they actually model a very large multi-agent simulation and then see that yes like the agents unfortunately whenever confronted with two distinct opposing beliefs like based upon the way they share their information and influence one another within the broader environment around them of which all other agents around them are part of the environment with which they interact receive observations commit actions such as sharing information um

they do start to form these polarizing groups so um per usual with modeling there are many other ways you could potentially approach that kind of question um this is just this was one of the the stronger uh papers that I've seen published and it's you know it's a rich field just as much as anything else um and so I I certainly went past what I intended to um but we can really quickly just give figure 3.3 a look um what this is and I'll try to do it quickly is to say um from left to right the left image is uh wi within you know we're modeling a two-dimensional random dynamical system with a non-equilibrium steady state what is that mean it means that we can have Dynamics such as noise um stochasticity it means with non-equilibrium that means that if there were some kind of steady state to be at that steady state is not entirely steady it can shift right um I mean my steady state at 2 am is to be asleep uh it is certainly not the case at 2 pm uh to be asleep and so that's one kind of crude way of saying that and then so with the left image this is where we have a system that actually does have a kind of surprise minimization Dynamic about it um they don't actually Define the specific rule it's following nor any of the variables the point is just to show that it tends towards uh starting from coordinates 55 it eventually tends towards over time this General Central area and doesn't quite leave there it doesn't necessarily have to stop at a particular point but it it stays generally around there and then the image to the far right we'll skip the middle one for a moment this is simply a kind of kind of gradient spread um or way of looking at what are the most surprising states that EX exist in this nine equilibrium uh steady state random dynamical system and so they say that this Center Circle is where states are least surprising you can kind of view this as there is an organism who's adapting and trying to learn how to live in this uh non-equilibrium steady state environment just like I have to figure out how to navigate the Dynamics of my body's need for sleep in ways that work well with say a kind of uh you know circadian clock where going to sleep roughly around the same time and getting up roughly around the same time every day uh tends to be most helpful uh both in practice and and and in study so um we could view it as that right uh it the view this as a kind of mirror the agent is kind of staying eventually arrives at and stays within this least surprising uh area of the the state space meanwhile the middle uh diagram is just to show we could have something that doesn't attempt to minimize surprise right it also starts at coordinate 55 but then it follows an incredibly kind of flu fluctuating uh you know irregular not very well patterned whatsoever path uh as it kind of succumbs to the the kind of stochastic uh you know dissipative uh entropic forces of the uh non-equil non-equilibrium steud State environment in which it exists and it because it's not you know there's no kind of control here uh we could phrase that in the

reinforcement learning sense for anyone familiar with that or simply the idea of you know a person who chooses to act against dis against entropy or not it's kind of like the left side is an example of uh uh an organiz striving to Exist by minimizing its surprise in order to instead stay in preferred non surprise inducing States uh as opposed to the middle graph where we end up with an agent who fully succumbs to to environmental Dynamics without any way of kind of putting up a a fight against them as it were we get many interesting relationships here as they uh the authors start to further tie together their work with with physics with information Theory and so on so we actually see that um the maximization of log probability of observations which is kind of just like saying um that we are trying to self-evidence we're seeking out observations that we want to see such as I want to not see my stomach growl I want to see my stomach silent and we could View that as in a sense being equivalent to the minimization of Shannon entropy for anyone familiar with Claud Shannon and his kind of representation of entropy in information Theory um but the the point is you know it's kind of kind of how spread out or uniform or uncertain is you know the the the the probability of the various kinds of evidence that we're finding we want to be able to find more certain evidence that makes us more certain cluding about our beliefs which in turn from The High Road of active inference would be very important for maintaining our survival I've certainly spoken for some time so I'm just curious if there's anyone who has any questions or or is curious about anything that they might want to discuss yeah so I I understand why you would model homeostasis as associ ating the preferred state with priors I I also see that there's evidence that very strong priors look like preferences in the world very often but going in the other direction is something I have problem with this idea so so going from very strong priors to preferences I think you can justify but going from but saying whenever there's a preference it's a very strong prior that that doesn't seem like it could possibly be adaptive because uh there there are uh you know there there are times when it's important to expect something that you don't want if you know it's typical for for example I think we talked about this John with the lion and the child a little bit so it is a really important question but the the kind of first principle here is it's a framework for expressing what you just said so you can go anywhere in the face space with this and that one precise and this and that one labile and this person's perspective on that one as that it just it's like in the generic case of generic homeostasis generalized homeostasis all that it's just expressivity and then how it is for any given thing capital t that is an empirical question or it's it's empirically analytical like just here's the properties of this map within this framework coordinate system so you know it's a valid concern but it's like there's no particular

life or death situation of any organism that is like necessarily like or the high road is at is at risk so would you say there are preferences that don't come from very strong priors that maybe all very strong priors act like preferences but maybe not all preferences come from very strong priors is that is that a an answer so just in terms of coherence with the uh the sort of ontology or the the terminology that's used in the textbook um so preferences here or we're looking at preferred States um they themselves could considered themselves to be to be priors um they're they're all kind of equivalent it depends on how you want to look at things and especially depends on how you want to model things so it's something that's very commonly done uh and I I kind of I don't know how well I described it earlier but um frequently we take priors over observations it's not always the case people can you know do this differently but it's very often the case that they they model prior specifically over observation rather than states to be representative of preferences um not even then not necessarily all cases um but and it maybe it would be best for me to jump to um a figure that could help kind of elucidate this a little bit better um a good example is the pp model um which will get to shortly but the the the point here we are so this is the PDP I think this features in chapter 7 um it and so here we're looking at this one this one parameter amidst everything else going on here um but C is is used to refer to uh priors over pre priors over observations AKA preferences and you you see that it's related to calculating G which is used to score policies what does that mean that means that you're able to score uh your potential actions available to you or or you're able to plan uh actions accordingly and engage in policy inference using uh these kind of expectations you have about what you are going to see which in terms of the letter c the parameter C is just a probability distribution priors over observations that stay fixed because they stay fix fixed which you can imagine like just like you know I need roughly 2,000 calories a day that's generally fixed whether I want to change my mind about that or not I will probably have a bad time if I try to change my mind about that because it will not align with you know what I repeatedly observe which is that you know I actually asked about that too for for exactly this reason so is maybe the resolution to the lion thing that the child starts with preferences he won't see any lion or starts with prior that he won't see any lions and then uh that initial prior is maintained as the preference even though the epistemological prior updates through basy and inference yeah that could be like a locked in state like I know there's a pro a project with PTSD Andre can share more but just those are the Dynamics big picture zooming out from this one specific issue about preference and prior and where's precise and whereas it isn't which system are we talking about the textbook is just saying here's the expressivity including like

nested systems systems with body and mind elements so it's like you got the grammar there you're asking the right formatting of question and you can actually just build maps of that to model different territories but that that's not on the high road that's the empirical half whereas this is more like your questions and thinking about thing Concepts yeah I'm just asking about this it's a chapter three and it's a number of chapters and it it's a bit weird like the pragmatic value thing it's associated with like I don't know if I I would say that pragmatic value is always you know a logarithm of probabilities it seems that that could work sometimes but maybe not all the time yeah it's I mean that's and that is exactly where preferences are are implicated but it's just I mean another way of viewing this idea of preferences and and I keep relating them to homeostasis and they even at one point in this chapter say that that preferences in a sense are like defining the phenotype of of an organism yeah well homostasis is is an example where this definitely work it's just yeah issues of having to predict the environment uh could could run you into trouble because uh if you if you act like you'll get what you want you you can get into trouble yeah I I I often whenever I I I use this uh you know caloric intake my stomach is growling therefore I'm hungry I use that example a lot just because it's so intuitive for literally everyone but I always I I sometimes extend it and say like well we're still in a partially observable environment what I mean is is my stomach growling a sign of hunger or is it a sign of like a throbbing stomach ulcer um the the you know the state I should you know infer you know if I'm not sure about one of those things or if I'm very sure that it's a sign of hunger when in fact it's a sign of a deeper problem I'm going to get very far like you know LED astray uh similarly with I I'm not totally familiar with this child and lion example that you all have given I'm guessing it's maybe a child who confronts uh you know a lion and his it's the point that the the child can't afford uh for the for his preferences about seeing a line to be similar to the to the actual probability belief because the the child has to not want to see the line and the child has to have an accurate probability of of the of running into it so so I mean we could even depends on how granular you want to go and and and Daniel did make the point about you know we end up making a lot of sort of modeling decisions even even in these kind of natural language examples that we give I mean another way you could you could drill into that even further and say even before the child makes any kind of like explicit knowing uh executive functioning uh kind of related uh inference about a lion um you could even you could even step further to the the fastest time scale which is like it might be that the child sees the line there's some kind of connection made there and suddenly the child uh you know starts to experience light amounts of perspiration a heightened heart rate um all kinds of other signs

of kind of physiological arousal that you know might lead one to infer that you know this child is now suddenly afraid yeah right and their stress system and cortisol and all of that have been activated U pretty rapidly so it's not just that we don't want to see a lion but like another way of viewing it as I don't want to see my body going go into that state um th this relates to uh uh kind of like a lot of work that's been done in affective Neuroscience so looking at like you know uh the idea of emotions actually being inferences from the physiological experiences we have um you know it's kind of like oh I'm angry I know I'm angry because I see that my you know my fist is shaking and I have a certain expression on my face you know almost unintentionally like a reaction so you can even relate you know phys the one's you know autonomic like fastpaced reactions as observations themselves that you don't want to see or do want to see um yeah I I hopefully I didn't like over complicate that or or make it more confusing but I hope that was useful in some kind of way um but it is a good it is a good question um another way of viewing preferences or or these kinds of expectations is you know later chapters again we look at like attracting points um the idea of you know these are kind of like attractors we we're already introduced to it here because we see in allor right figure yeah that we spent some time with already like this leftmost area this is like you know following a system and and moving towards preferred States or moving towards like this this kind of preferred region um as opposed to not having that with so so so it's kind of like on the left it's like with these kinds of surprise minimization Dynamics involved there's a tendency to move a certain way um as opposed to in the Middle where you know there's no attractor here there's there's no this is pure kind of FL depending on maybe your initial starting point and then whatever plays out from there in you know your error terms and you know whatever else is involved but there's nothing pulling you towards you know say some low surprisal region of a state space um you know it' be like again me not eating food and slowly succumbing to the you know the physical processes around me um and another interesting thing for those interested in computational modeling is also uh related which is how what happens you know we just looked at that middle graph of like you know the the the agent who's kind of spiraling out of nowhere there's no attractor um they did another experiment when I say they this experiment goes pretty far back or at least not in 2022 when this textbook was published um but uh they it's this um there's this phenomena um oh 1985 uh phenomena of the inhibition of return which is pretty interesting so um John you previously brought up the idea of pragmatic value and we know from uh previously I I don't want this equation to mislead us maybe I'll jump back um because this one has to do with with using different kinds of terms including

novelty um which also is a fair breakdown of expected free energy but still I just want to stick to um differentiating epistemic from equations of course um differentiating epistemic value AKA Information Gain with pragmatic value which as you saw whenever this little icon guy isn't there here we go um see the preferences are implicating the pragmatic value it's kind of like you know um I could eat food or I could spend some time looking up restaurants of where I could get food um yeah if I spent time looking up restaurants maybe I'll end up learning something interesting uh and but it's the actual Act of like attaining and eating the food that that kind of bestows the direct pragmatic value of realizing my preferences here in the sense of not wanting to see my my stomach Rowling living along those lines um so it's only in C that this plays out right so that kind of means information game Information Gain in one sense okay so this the one problem I'm having is I don't really understand what C is okay sure so um one way we could look at C is this same chapter here with the PDP um here we are so I don't mean to skip ahead so much and I don't want to overwhelm but you can see here that this is for the tze problem I believe yeah um this is the tze problem and let's just ignore the these are two different submatrices within the big C sub uh full Matrix containing preferences prior over observations so in the tze the the mouse can receive it has two different observation modalities we're not going to over complicate we're just going to focus on one of them it's the second one here what does this second one represent well we have essentially a single Vector right so it's a vector is you know like a like a one column or one row um transposed Matrix so the first entry is zero which based on the problem I'll just tell you what it is it is um this is to not see um a reward or nor a shock the here's the t- maze the t- maze is you can uh the mouse can essentially it can go uh to multiple spaces within this little t-shaped maze it can go left uh or it go right meaning it can go take a left wing or right wing based on its starting position and kind of closer to the center down uh area and and the idea is on the left will be either a piece of cheese like a reward um and on the other side will be a negative stimulus such as like a little shock or something along those lines but you can you can already make out the the abstract notion that is you know this is an opportunity to achieve pragmatic value uh by going for you know cheese and IDE ideally the the mouse wants to to find the cheese and so it's going to have to engage in state inference about trying to figure out which the left or right arm of the teamas like which location uh the the the C1 relates to locations and observing the locations it's at um C2 is it this this first entry is where um there's nothing meaning no cheese but also no negative stimulus it has a preference of zero and we're just just with me about what these numbers mean they're a little arbitrary but they they mean something um this middle value six

is saying that with with kind of a power of or or number of six the the mouse prefers the cheese right the the reward um this the second entry represents the the probability for the reward um and then the third one negative -6 obviously very bad this is you can already see here this is their preference for the aversive stim with it being negative negative being less than zero you you can see it's almost like a ranking system here of the the mouse strongly dis prefers the aversive stimulus um it is pretty indifferent neutral to the absence of either the reward or the negative stimulus and then it strongly prefers the observation of the cheese right what the reason why this doesn't look like a probability distribution is simply that we you know you could Define it as a probability distribution and that is ultimately what it is um sometimes people in active inference they do it this way just to make it more intuitive um but in any case you run it through the softmax function right and whenever you do that that is what kind of converts it into a probability distribution but then furthermore it amplifies um the the resulting probability distribution so what what's going to happen is you know it it just look at how it's for practically this formulation means the attractive stimulus is considered e to six or 400 times more probable than the neutral stimulus under the rat's generative model so kind of just you the softmax comes up often it also comes up in Precision modulation um but um sorry I just realized uh Daniel your hand was up yeah I just wanted to add one point on that the the soft Max point so that sums an arbitrary set of numbers to one so it's like this and and this is the kind of stuff that gets learned in stats classes like if you don't sum it to one or frame it in terms of a distribution you're dealing with a scenario where like even on the next time step there's either less than something happening or more than something happening like you C you can't sum up the events that could happen for a given variable it's like that has to sum to one to reflect the Total distribution of what could happen for that to persist so it's like to go off of that kind of summing to one statistical approach um it it's like it puts you outside of the empirical because you want to have a framework where you can observe it and model it statistically otherwise it's kind of like saying like there's some other way to model noisy processes or processes with partial information that doesn't fall under sta statistical approach wait so to maintain the sum of one is there some kind of unitary Evolution yeah like in the quantum setting that is called the unitary and at a deeper level but just at a more practical level the softmax takes this string like 066 or 011 or 1 z0 or2 and it proportional lies it and you can adjust the the posterior proportioning like the pie chart that is the soft Max output of these numbers like that's 100% tunable the question which numbers to use so after you do the soft backs you still have a vector it's just that all of these sum to one and the the higher values are

Amplified yeah like just statistical distributions overall like gaussian with a mean and a variance it's it's two numbers but because you know that it's going to get plugged into a gaussian you know that it's going to sum to one under the curve can this change over time these numbers yes preferences can be updated any preference in a given model any any variable could be fixed or learnable okay so yeah so you you you you yeah that that'll give you unitary Evolution so if you evolve it with a matrix that Matrix will have to be unitary so yeah it's kind of like y equals 4 like it's a function that on one hand it does it does have time as a dimension it could but it just stays at four so it's like whether you choose a fixed parameter or you choose like kind of like a colan filter like a distribution unfolding through time with it just you're within the statistics dummy that's what parameter updating and learning is about using statistical methods so it's like again if you have some other method this is the what this is what the current statistical packages are implementable it's not that there aren't methods outside of doing probabilistic statistics and there's yeah yeah it I mean with with the examples that we see often in like kind of the textbook oriented um examples that are given the teamm ETC um the preferences are not learned but you know seconding Daniel they absolutely could be learned which is part of why I brought up like this paper here among you know probably a dozen others at this point but yeah any parameter any of these you know probability distribution within your agent can be learned and it can be learned via free energy minimization essentially right um it's only policy inference that that uses expected free energy so um yeah everything else is updatable the the thing is it's kind of tough because you know with the idea of homeostasis or having preferred States it's like you would assume that those wouldn't really change too much um we can think of many instances where they would change I mean you can think of you know a child at a very young age requires a significantly lower caloric intake than like you know a 20 or 30-year-old adult and then even further once they're at an older age the the required caloric intake starts to go down again so I mean that that's one way of viewing it it's like is there preference learning is there just genuine you know changes within the organism such that what defines its homeostasis itself changes over time um there are different ways you can look at so so that's like a soft natural language explanation that refers to physiology meanwhile you can also just look at it as purely like you know statistics and machine learning or do you want this model to be learned or learnable or not um do you want to parameterize it in a particular way you know what it what is you know anything related to things like like learning rates or or precisions or uh modulation or otherwise all those things could apply um it's really just what do you want to model what do you want to

create and what actually makes sense and then you know ideally if possible depending on your phenomena of interest you could engage from there and like empirical testing and and fitting of the model to empirical data if you're in that kind of domain that you know that that something like that would make sense you know um I mean similarly you could the you know so so so D meanwhile you notice how similar these look to see except they don't do the soft Max instead they're just at the start they're making it clear that this is going to sum to one this sums to one they I don't know entirely why they do that I don't think it makes it that much more intuitive to understand but uh to to multiply this by one half but you can imagine this is 0.5 like this is a 50/50 prior over hidden States whereas C was a prior over the observations such as is there cheese here is there a negative stimuli here is there nothing uh you know absence of those things here um here it's like you know D is let's see D2 two contexts oh is the cheese on the left or is it on the right you know starting with a prior of which is essentially after we multiply uh you know 50/50 it's 0.5 versus 0.5 something to one know so um yeah and then I guess just well we're about out of time I just I really like this uh this inhibition of return concept it's simply if you if you recall with with expected free energy minimization we have Information Gain and pragmatic value you can imagine if C was highly uniform you know like uh you let's say the the the mouse equally prefers um nothing as it prefers like imagine all three of these ignore the soft max or you could still leave it there either way let's imagine these were 33.33 .33 you know essentially summing to one uh and and so there's an equivalent preference for everything right meaning no it's a fully uniform distribution you could plot out the pmf just you know everything's at the same level um and what would happen is your expected free energy term it'd be very difficult right to to rank available actions to take should I go left should I go right should I stop for a second think about it I'm not sure because I don't know how to weigh any of these in terms of pragmatic value because I don't know how to derive PR pragmatic value because I don't prefer any one particular thing versus another it's all equivalent well then what happens in that kind of strange hypothetical case you're only relying on Information Gain whenever it comes to action selection and policy inference so what does that mean that means that all of your actions since since there's no reward you know nothing is can be defined as reward you don't prefer anything more than anything else so instead all of your behavior is essentially based on sorry I'm scrolling around so much I just um but uh all of your behavior will be centered on Information Gain in which case they they did this um let's see Yep this experiment where they just have an agent navigate like a a maze or is it even a maze is it just a grid um like there's nothing to be you know

solved there's no maze to be conquered or won because you did the correct thing because you don't you don't care essentially you don't care because your your your beliefs are are uniform um and so they just the whole point is this agent who has no ability to figure out some kind of like you know Tipping Point for pragmatic value because they prefer everything equally they end up just moving around the maze but in interestingly they don't stay why don't they stay put they don't because there's no information gain from staying put right so it's just kind of to show like there there's this kind of balancing actor I don't even know if that's the right way to phrase it but we have Information Gain have pragmatic value these are the two main things that kind of Define behavior it's like how do I choose what to do well will I get new information by choosing an action or will I get pragmatic value out of choosing an action or will I get both you know there's no they're not not necessarily two contending or fighting uh values right it's simply that you can compute both because we you know essentially have the sort of equations for that where of course we have X you know our states our observations and our actions the three essential components of model as well as C and then again the degree of uniformity of C can very much um impact the behavior of the agent anyway I recognize we've gone over on time um but um interesting conversation thank you for your questions John appreciate it I'm sure they they'll be useful to to other people who uh watch that this if not now then in future on a on the recording so um thank you everyone and then next week um at this same time it'll be back to cohort 7 and we'll be looking at chapter nine uh of the textbook that looks at you know data analysis essentially and model analysis and and just like you know group comparisons and a lot of kind of interesting things that happen after you have defined your model and your environments and you kind of start playing things out uh and then that morning we'll we'll have cohort 8 I I think we continuing with the same chapter if I'm not mistaken so um thank you all so much and uh have a great evening or whatever time it is for you now so have a good one thank you bye